

# **“Collaborative Story/Screenplay Writing Platform with Version History”**

## **CO4 ASSESSMENT TOOL 1**

*Submitted in the partial fulfilment for the Course of*

### **CSA1016 - Software Engineering**

*to the award of the degree of*

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**ENGINEERING IN**

**Artificial Intelligence & Data Science**

**Submitted by**

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**SIMATS**  
**ENGINEERING**



**SIMATS**  
Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

**SIMATS ENGINEERING**

**Saveetha Institute of Medical and Technical**

**Sciences Chennai-602105**

**August 2025**



# **SIMATS ENGINEERING**

**Saveetha Institute of Medical and Technical  
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## **DECLARATION**

I, **Lithish Kumar R 192424179**, of the **Department of Artificial Intelligence and Data Science**, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled “**Collaborative Story/Screenplay Writing Platform with Version History**” is the result of our own Bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: Chennai  
Date: 14/08/2026

Signature of the Students

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## 1. Problem Statement

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The assigned problem statement is a Collaborative Story / Screenplay Writing Platform with Version History. The platform enables multiple authors, co-writers and reviewers to create, edit and manage screenplay or story documents together in real time. Every meaningful change is captured as a version snapshot so that authors can review the evolution of a script, compare drafts, and revert to an earlier version when required. The platform must support role-based collaboration (Owner, Editor, Commenter, Viewer), inline commenting, conflict detection when two users edit the same content simultaneously, a formal review/publish workflow, and export to industry-standard screenplay formats such as PDF, Final Draft (.fdx) and Fountain.

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This document designs and documents a comprehensive suite of test cases to verify the correctness, functionality, reliability, security and usability of the platform, in line with the CO4 rubric criteria: requirement/

**scenario identification, test case completeness, test data and expected results, application of test design techniques, traceability and coverage, and overall documentation quality.**

## 2. Requirement Identification

### 2.1 Functional Requirements (FR)

| ID   | Functional Requirement                                                                                         |
|------|----------------------------------------------------------------------------------------------------------------|
| FR1  | The system shall allow a user to register and authenticate using email/password or SSO.                        |
| FR2  | The system shall allow an authenticated user to create a new story/screenplay document.                        |
| FR3  | The system shall allow multiple users to edit the same document simultaneously in real time.                   |
| FR4  | The system shall automatically create a version snapshot at defined intervals and on explicit save.            |
| FR5  | The system shall allow a user to view the complete version history of a document.                              |
| FR6  | The system shall allow an authorized user to restore/revert a document to any previous version.                |
| FR7  | The system shall allow the document owner to assign roles (Owner, Editor, Commenter, Viewer) to collaborators. |
| FR8  | The system shall restrict actions (edit, delete, invite, publish) based on the collaborator's assigned role.   |
| FR9  | The system shall detect simultaneous conflicting edits and provide a merge/conflict-resolution mechanism.      |
| FR10 | The system shall allow users to add inline comments and suggestions on specific lines or scenes.               |
| FR11 | The system shall allow export of the screenplay in PDF, Final Draft (.fdx) and Fountain formats.               |
| FR12 | The system shall send email/in-app notifications for invites, comments and version changes.                    |
| FR13 | The system shall allow a document to move through workflow states: Draft, Under Review, Published, Archived.   |
| FR14 | The system shall allow keyword search within a document and across a user's document library.                  |

### 2.2 Non-Functional Requirements (NFR)

| ID   | Quality Attribute | Requirement                                                                                         |
|------|-------------------|-----------------------------------------------------------------------------------------------------|
| NFR1 | Performance       | Collaborative edits shall be reflected to all connected users within 500 ms.                        |
| NFR2 | Scalability       | The system shall support at least 50 concurrent collaborators on a single document.                 |
| NFR3 | Reliability       | The system shall provide a minimum of 99.5% uptime availability.                                    |
| NFR4 | Security          | Data in transit shall use TLS 1.2+ and data at rest shall be encrypted with AES-256.                |
| NFR5 | Usability         | A first-time user shall create and share a document within 2 minutes without training.              |
| NFR6 | Data Integrity    | No version snapshot shall be lost, duplicated or corrupted during auto-save.                        |
| NFR7 | Compatibility     | The system shall function correctly on the latest two versions of Chrome, Firefox, Edge and Safari. |
| NFR8 | Auditability      | All role changes and version restores shall be logged with timestamp and user ID.                   |
| NFR9 | Recoverability    | Unsaved edits shall be recovered within 5 seconds of reconnect after a network drop.                |

### 2.3 Test Scenarios

**Test scenarios are grouped into nine functional modules to ensure comprehensive requirement and functional coverage across the platform.**

#### **Module A: User Authentication & Registration**

- TS1: Verify user registration with valid and invalid details
- TS2: Verify login with valid, invalid and locked-out credentials

**Module B: Document Creation & Real-Time Editing**

- TS3: Verify creation of a new story/screenplay document
- TS4: Verify real-time collaborative editing between two or more users
- TS5: Verify automatic and manual save/version snapshot creation

**Module C: Version History Management**

- TS6: Verify listing and chronological ordering of version history
- TS7: Verify restoring a document to a previous version
- TS8: Verify side-by-side comparison (diff) between two versions

**Module D: Roles & Permission Management**

- TS9: Verify owner can assign/change collaborator roles
- TS10: Verify an Editor cannot perform Owner-only actions
- TS11: Verify a Viewer cannot edit or comment on the document

**Module E: Conflict Detection & Resolution**

- TS12: Verify conflict is detected when two users edit the same line simultaneously
- TS13: Verify the merge/conflict-resolution UI allows correct resolution

**Module F: Commenting & Suggestions**

- TS14: Verify inline comments can be added, edited, resolved and deleted

**Module G: Export & Interoperability**

- TS15: Verify export of a screenplay to PDF with correct formatting
- TS16: Verify export to Final Draft (.fdx) and Fountain formats

**Module H: Workflow State Transitions**

- TS17: Verify document transitions correctly across Draft, Under Review, Published, Archived and Rejected states

**Module I: Notifications**

- TS18: Verify notifications are triggered for invites, comments and version restores

**Module J: Non-Functional & Security Testing**

- TS19: Verify performance, scalability, security, compatibility and recoverability requirements (NFR1-NFR9)

### 3. Detailed Test Cases

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Each module below documents detailed test cases covering normal, boundary, invalid and exceptional conditions, with explicit Test Case ID, Test Scenario reference, Test Steps, Test Data, Expected Result, Actual Result and Pass/Fail Status. Actual Result and Status assume a defect-free build at time of execution and would be updated during real test-cycle execution.

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#### 3.1 Module A - User Authentication & Registration

| Test Case ID | Test Scenario | Test Steps                                                                                                      | Test Data                                                    | Expected Result                                                    | Actual Result | Status |
|--------------|---------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------|---------------|--------|
| TC_AUT_H_01  | TS1 - Normal  | Enter valid name, valid email, password meeting policy (8+ chars, 1 upper, 1 digit, 1 special); click Register. | Name: Ariya Sen;<br>Email: ariya.sen@mail.com; Pwd: Pass@123 | Account created; verification email sent; redirected to dashboard. | As Expected   | Pass   |

| Test Case ID | Test Scenario     | Test Steps                                                                                     | Test Data                                     | Expected Result                                                                                          | Actual Result | Status |
|--------------|-------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------|--------|
| TC_AUT_H_02  | TS1 - Invalid     | Enter an email address already registered in the system; click Register.                       | Email: existing.user@mail.com                 | System rejects with 'Email already registered' message; no duplicate account created.                    | As Expected   | Pass   |
| TC_AUT_H_03  | TS1 - Boundary    | Enter password exactly at minimum boundary of 8 characters and one below (7 characters).       | Pwd1: Ab1@abcd (8); Pwd2: Ab1@abc (7)         | 8-char password accepted; 7-char password rejected with policy error.                                    | As Expected   | Pass   |
| TC_AUT_H_04  | TS2 - Normal      | Enter registered email and correct password; click Login.                                      | Email: ariya.sen@mail.com; Pwd: Pass@123      | User authenticated and redirected to the document library.                                               | As Expected   | Pass   |
| TC_AUT_H_05  | TS2 - Exceptional | Enter wrong password 5 consecutive times.                                                      | Pwd attempts: 5 incorrect passwords           | Account temporarily locked for 15 minutes after the 5th failed attempt; lockout notice shown.            | As Expected   | Pass   |
| TC_AUT_H_06  | TS2 - Normal      | Click 'Sign in with Google/SSO' and complete the provider's consent flow with a valid account. | SSO provider: Google; valid corporate account | User is authenticated via SSO and redirected to the document library without a separate password prompt. | As Expected   | Pass   |

### 3.2 Module B - Document Creation & Real-Time Editing

| Test Case ID | Test Scenario  | Test Steps                                                                                   | Test Data                                                 | Expected Result                                                                       | Actual Result | Status |
|--------------|----------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------|---------------|--------|
| TC_DOC_01    | TS3 - Normal   | Click 'New Document', select 'Screenplay' template, enter a title, click Create.             | Title: 'The Last Reel' (24 chars)                         | New screenplay document created and opened in the editor within 2 seconds.            | As Expected   | Pass   |
| TC_DOC_02    | TS3 - Invalid  | Attempt to create a document leaving the Title field empty.                                  | Title: " (0 characters)                                   | System blocks creation and shows 'Title is required' validation message.              | As Expected   | Pass   |
| TC_DOC_03    | TS3 - Boundary | Enter a title of exactly 100 characters and then 101 characters.                             | Title1: 100 chars; Title2: 101 chars                      | 100-char title accepted; 101-char title rejected with 'Max 100 characters' error.     | As Expected   | Pass   |
| TC_DOC_04    | TS4 - Normal   | Two users (User A, User B) open the same document and type in different scenes concurrently. | User A edits Scene 1; User B edits Scene 3 simultaneously | Both users' edits appear on-screen for each other within 500 ms; no data loss.        | As Expected   | Pass   |
| TC_DOC_05    | TS5 - Normal   | Type continuously for 3 minutes without manual save; observe auto-save indicator.            | Auto-save interval: every 60 seconds                      | A new version snapshot is auto-generated every 60 seconds; 'Saved' indicator updates. | As Expected   | Pass   |

| Test Case ID | Test Scenario       | Test Steps                                                                                                       | Test Data                          | Expected Result                                                                          | Actual Result | Status |
|--------------|---------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------|------------------------------------------------------------------------------------------|---------------|--------|
| TC_DOC_06    | TS5 - Exceptional   | Disconnect network for 30 seconds while editing, then reconnect.                                                 | Simulated network drop: 30 sec     | Unsaved edits are cached locally and synced automatically within 5 seconds of reconnect. | As Expected   | Pass   |
| TC_DOC_07    | TS3 - Normal (FR14) | Use the global search bar to search for a keyword that appears in two different documents in the user's library. | Search term: 'reel' across library | Both matching documents are returned with the matching line snippet highlighted.         | As Expected   | Pass   |

### 3.3 Module C - Version History Management

| Test Case ID | Test Scenario     | Test Steps                                                                                                                 | Test Data                    | Expected Result                                                                                   | Actual Result | Status |
|--------------|-------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------------------------------------------------------|---------------|--------|
| TC_VER_01    | TS6 - Normal      | Open the 'Version History' panel of a document with 6 saved versions.                                                      | Document with versions v1-v6 | All 6 versions listed in reverse-chronological order with timestamp and author.                   | As Expected   | Pass   |
| TC_VER_02    | TS6 - Boundary    | Open version history for a brand-new document (0 prior versions) and for a document at the max stored limit (50 versions). | 0 versions; 50 versions      | 0-version doc shows only the current draft; 50-version doc lists all 50 without truncation error. | As Expected   | Pass   |
| TC_VER_03    | TS7 - Normal      | Select version v3 from history and click 'Restore'.                                                                        | Target version: v3 of 6      | Document content reverts to v3; a new version v7 is created preserving history (no overwrite).    | As Expected   | Pass   |
| TC_VER_04    | TS7 - Invalid     | Attempt to restore a version ID that does not exist (tampered request).                                                    | Version ID: v999 (invalid)   | System returns a 404/'Version not found' error; no change to the document.                        | As Expected   | Pass   |
| TC_VER_05    | TS8 - Normal      | Select v2 and v5, click 'Compare'.                                                                                         | Compare v2 vs v5             | Diff view highlights additions in green and deletions in red, line by line.                       | As Expected   | Pass   |
| TC_VER_06    | TS8 - Exceptional | Attempt to compare a version against itself.                                                                               | Compare v4 vs v4             | System shows 'No differences found' message instead of an empty/broken diff.                      | As Expected   | Pass   |

### 3.4 Module D - Roles & Permission Management

| Test Case ID | Test Scenario | Test Steps                                                                                   | Test Data                             | Expected Result                                                                       | Actual Result | Status |
|--------------|---------------|----------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------|---------------|--------|
| TC_ROLE_01   | TS9 - Normal  | Owner opens Share settings, enters collaborator email, selects role 'Editor', clicks Invite. | Role: Editor; Email: collab1@mail.com | Invitation sent; collaborator appears as 'Pending - Editor' in the collaborator list. | As Expected   | Pass   |

| Test Case ID | Test Scenario  | Test Steps                                                                            | Test Data                               | Expected Result                                                                          | Actual Result | Status |
|--------------|----------------|---------------------------------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------|---------------|--------|
| TC_ROLE_02   | TS9 - Boundary | Owner adds the 50th and then the 51st collaborator to a document.                     | Collaborator count: 50 then 51          | 50th collaborator added successfully; 51st is blocked with 'Collaborator limit reached'. | As Expected   | Pass   |
| TC_ROLE_03   | TS10 - Invalid | Logged in as Editor, attempt to delete the document or change another user's role.    | Role: Editor attempting Owner action    | Action is blocked; 'Insufficient permissions' message shown; document unaffected.        | As Expected   | Pass   |
| TC_ROLE_04   | TS11 - Invalid | Logged in as Viewer, attempt to type in the editor or add a comment.                  | Role: Viewer                            | Editor is in read-only mode; comment box is disabled/hidden for Viewer role.             | As Expected   | Pass   |
| TC_ROLE_05   | TS11 - Normal  | Logged in as Viewer, verify the document content renders correctly in read-only mode. | Role: Viewer, Document: 'The Last Reel' | Full document content is visible and scrollable; no edit cursor is available.            | As Expected   | Pass   |

### 3.5 Module E - Conflict Detection & Resolution

| Test Case ID | Test Scenario      | Test Steps                                                                                                   | Test Data                                         | Expected Result                                                                            | Actual Result | Status |
|--------------|--------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------------------------------------------------|---------------|--------|
| TC_CON F_01  | TS12 - Normal      | User A and User B edit the exact same line/word at the same time.                                            | Line 12: both users edit simultaneously           | System detects the conflict and flags the line with a conflict indicator for both users.   | As Expected   | Pass   |
| TC_CON F_02  | TS12 - Boundary    | Two users edit adjacent but non-overlapping words in the same line at the same time.                         | Line 12: User A edits word 1, User B edits word 5 | No conflict is raised; both edits are merged automatically using operational transforms.   | As Expected   | Pass   |
| TC_CON F_03  | TS13 - Normal      | On conflict, open the resolution dialog and choose 'Keep mine' for one change and 'Keep theirs' for another. | 2 conflicting edits                               | Selected versions are applied correctly; resolved line matches the chosen selections.      | As Expected   | Pass   |
| TC_CON F_04  | TS13 - Exceptional | Both users close the conflict dialog without resolving and one disconnects.                                  | Unresolved conflict + disconnect                  | System auto-saves the last-known-good state and re-opens the conflict prompt on reconnect. | As Expected   | Pass   |
| TC_CON F_05  | TS12 - Exceptional | A third user attempts to export the document while an unresolved conflict exists on another line.            | Export request during open conflict               | Export is blocked or clearly flagged as based on the last conflict-free saved version.     | As Expected   | Pass   |

### 3.6 Module F - Commenting & Suggestions

| Test Case ID | Test Scenario   | Test Steps                                                                                                 | Test Data                                         | Expected Result                                                                        | Actual Result | Status |
|--------------|-----------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------|---------------|--------|
| TC_CMT_01    | TS14 - Normal   | Select a line of dialogue, click 'Add Comment', type feedback, click Post.                                 | Comment: 'Tighten this exchange' (24 chars)       | Comment appears anchored to the selected line with author name and timestamp.          | As Expected   | Pass   |
| TC_CMT_02    | TS14 - Boundary | Enter a comment of exactly 500 characters (max limit) and 501 characters.                                  | Comment1: 500 chars; Comment2: 501 chars          | 500-char comment posts successfully; 501-char comment is blocked with a limit warning. | As Expected   | Pass   |
| TC_CMT_03    | TS14 - Normal   | Mark an existing comment thread as 'Resolved'.                                                             | Existing comment thread ID: C-102                 | Thread is greyed out/collapsed and moved to the 'Resolved' filter tab.                 | As Expected   | Pass   |
| TC_CMT_04    | TS14 - Invalid  | Attempt to delete another user's comment while logged in as a Commenter (not the comment author or Owner). | Role: Commenter deleting another author's comment | Delete option is unavailable/blocked; only the comment author or Owner may delete it.  | As Expected   | Pass   |

### 3.7 Module G - Export & Interoperability

| Test Case ID | Test Scenario      | Test Steps                                                                                                | Test Data                      | Expected Result                                                                                     | Actual Result | Status |
|--------------|--------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------|---------------|--------|
| TC_EXP_01    | TS15 - Normal      | Open a completed screenplay, click Export > PDF.                                                          | Document: 90-page screenplay   | PDF is generated with correct scene headings, industry-standard margins and page numbers.           | As Expected   | Pass   |
| TC_EXP_02    | TS15 - Boundary    | Export a document with 0 scenes (empty) and one with 500 scenes (very large).                             | 0 scenes; 500 scenes           | Empty doc shows 'Nothing to export' warning; 500-scene doc exports without truncation or timeout.   | As Expected   | Pass   |
| TC_EXP_03    | TS16 - Normal      | Export the same document as .fdx and as Fountain (.fountain).                                             | Formats: .fdx, .fountain       | Both files are generated, downloadable, and open correctly in third-party screenwriting tools.      | As Expected   | Pass   |
| TC_EXP_04    | TS16 - Exceptional | Attempt export while collaborative edits are actively in progress by another user.                        | Export during live edit        | Export uses the latest saved snapshot; a notice indicates unsaved live changes are excluded.        | As Expected   | Pass   |
| TC_EXP_05    | TS15 - Boundary    | Export a document whose title and scene headings contain Unicode/special characters and accented letters. | Title: 'Café Noir — Scene 1/2' | PDF/FDX/Fountain files are generated with all special characters rendered correctly, no corruption. | As Expected   | Pass   |

### 3.8 Module H - Workflow State Transitions

| Test Case ID | Test Scenario      | Test Steps                                                                                                         | Test Data                               | Expected Result                                                                                | Actual Result | Status |
|--------------|--------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------|---------------|--------|
| TC_STAT E_01 | TS17 - Normal      | From Draft, click 'Submit for Review'.                                                                             | Current state: Draft                    | State changes to 'Under Review'; reviewers are notified.                                       | As Expected   | Pass   |
| TC_STAT E_02 | TS17 - Normal      | Reviewer clicks 'Approve' on an Under Review document.                                                             | Current state: Under Review             | State changes to 'Published'; document becomes read-only for Editors.                          | As Expected   | Pass   |
| TC_STAT E_03 | TS17 - Normal      | Reviewer clicks 'Reject' with comments on an Under Review document.                                                | Current state: Under Review             | State changes to 'Rejected'; author is notified with reviewer comments.                        | As Expected   | Pass   |
| TC_STAT E_04 | TS17 - Invalid     | Attempt to directly transition a document from 'Draft' to 'Published' via a crafted API request, bypassing review. | Invalid transition: Draft -> Published  | System rejects the invalid state transition and logs the attempt; state remains 'Draft'.       | As Expected   | Pass   |
| TC_STAT E_05 | TS17 - Exceptional | Restore an 'Archived' document.                                                                                    | Current state: Archived                 | Document returns to 'Draft' state as a new version branch; original archive entry is retained. | As Expected   | Pass   |
| TC_STAT E_06 | TS17 - Normal      | From the document library, filter the list view by state = 'Archived'.                                             | Filter: Archived, 12 archived documents | Only the 12 archived documents are displayed, each tagged with its archive date.               | As Expected   | Pass   |

### 3.9 Module I - Notifications

| Test Case ID | Test Scenario      | Test Steps                                                                                    | Test Data                        | Expected Result                                                                                       | Actual Result | Status |
|--------------|--------------------|-----------------------------------------------------------------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------|---------------|--------|
| TC_NOTI F_01 | TS18 - Normal      | Owner invites a new collaborator to a document.                                               | Invite sent to collab2@mail.com  | Collaborator receives an email and in-app notification within 1 minute.                               | As Expected   | Pass   |
| TC_NOTI F_02 | TS18 - Normal      | A collaborator restores an older version of a shared document.                                | Version restore action by Editor | All other active collaborators receive an in-app notification identifying who restored which version. | As Expected   | Pass   |
| TC_NOTI F_03 | TS18 - Exceptional | Trigger 20 notification events for a single user within 1 minute.                             | 20 rapid comment notifications   | Notifications are intelligently batched into a single digest instead of 20 separate alerts.           | As Expected   | Pass   |
| TC_NOTI F_04 | TS18 - Boundary    | Disable all in-app notifications in user preferences, then have a collaborator add a comment. | Notification preference: OFF     | No in-app alert is shown; the comment still appears in-document and in the activity log.              | As Expected   | Pass   |

### 3.10 Module J - Non-Functional & Security Testing

| Test Case ID | Test Scenario                | Test Steps                                                                                                               | Test Data                                | Expected Result                                                                                     | Actual Result | Status |
|--------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------|--------|
| TC_NFR_01    | TS19 - Performance (NFR1)    | Two users on separate sessions type simultaneously in the same document while latency is measured via browser dev tools. | 50 keystrokes/user over 30 sec           | Edits are reflected on the peer's screen within 500 ms for at least 95% of keystrokes.              | As Expected   | Pass   |
| TC_NFR_02    | TS19 - Scalability (NFR2)    | Load-test the document with a scripted 50 simulated concurrent collaborators editing different sections.                 | 50 concurrent virtual users (JMeter)     | System remains responsive with no dropped edits and CPU/memory within acceptable thresholds.        | As Expected   | Pass   |
| TC_NFR_03    | TS19 - Security (NFR4)       | Intercept network traffic during login and editing using a proxy tool; inspect the transport protocol.                   | Traffic capture during active session    | All traffic is encrypted via TLS 1.2+; no credentials or document content are visible in plaintext. | As Expected   | Pass   |
| TC_NFR_04    | TS19 - Compatibility (NFR7)  | Execute the core editing and export workflow on the latest two versions of Chrome, Firefox, Edge and Safari.             | 4 browsers x 2 versions = 8 combinations | All core workflows complete successfully with consistent layout across every combination.           | As Expected   | Pass   |
| TC_NFR_05    | TS19 - Recoverability (NFR9) | Begin typing, forcibly disconnect the network for 10 seconds mid-sentence, then restore connectivity.                    | Simulated 10-second network outage       | Local edits are queued and auto-synced within 5 seconds of reconnect with no data loss.             | As Expected   | Pass   |
| TC_NFR_06    | TS19 - Auditability (NFR8)   | Change a collaborator's role from Editor to Viewer and restore an old version; inspect the audit log.                    | Role change + version restore actions    | Both actions appear in the audit log with correct user ID, action type and timestamp.               | As Expected   | Pass   |

## 4. Application of Test Design Techniques

### 4.1 Equivalence Partitioning (EP)

Input domains are partitioned into valid and invalid classes so that one representative test value from each class provides confidence for the entire class, reducing redundant test cases while retaining coverage.

| Input Field       | Partition Type | Range / Class                           | Representative Value | Expected Outcome                |
|-------------------|----------------|-----------------------------------------|----------------------|---------------------------------|
| Document Title    | Valid          | 1-100 characters                        | "The Last Reel"      | Accepted                        |
| Document Title    | Invalid (Low)  | 0 characters (empty)                    | ""                   | Rejected - required field error |
| Document Title    | Invalid (High) | > 100 characters                        | 101-char string      | Rejected - max length error     |
| Password          | Valid          | 8-64 chars, mixed case + digit + symbol | Pass@1234            | Accepted                        |
| Password          | Invalid        | < 8 characters or missing symbol/digit  | abcdefg              | Rejected - policy error         |
| Collaborator Role | Valid          | {Owner, Editor, Commenter, Viewer}      | Editor               | Role assigned                   |
| Collaborator Role | Invalid        | Any value outside defined set           | "SuperAdmin"         | Rejected - invalid role error   |

### 4.2 Boundary Value Analysis (BVA)

**Defects are statistically more likely to occur at the edges of input ranges. BVA tests values immediately below, at, and immediately above each documented limit.**

| Field / Limit                                | Boundary Under Test | Test Values        | Expected Result                                              |
|----------------------------------------------|---------------------|--------------------|--------------------------------------------------------------|
| Document Title length (max 100)              | Min/Max boundary    | 0, 1, 99, 100, 101 | 0 & 101 rejected; 1, 99, 100 accepted                        |
| Collaborators per document (max 50)          | Upper boundary      | 49, 50, 51         | 49 & 50 accepted; 51 rejected with limit message             |
| Comment length (max 500)                     | Upper boundary      | 499, 500, 501      | 499 & 500 accepted; 501 rejected                             |
| Auto-save interval (30-120 sec configurable) | Min/Max boundary    | 29, 30, 120, 121   | 29 & 121 rejected as invalid config; 30 & 120 accepted       |
| Version history retained (max 50 versions)   | Upper boundary      | 49, 50, 51         | 51st version triggers oldest-version archival, not data loss |

### 4.3 Decision Table Testing

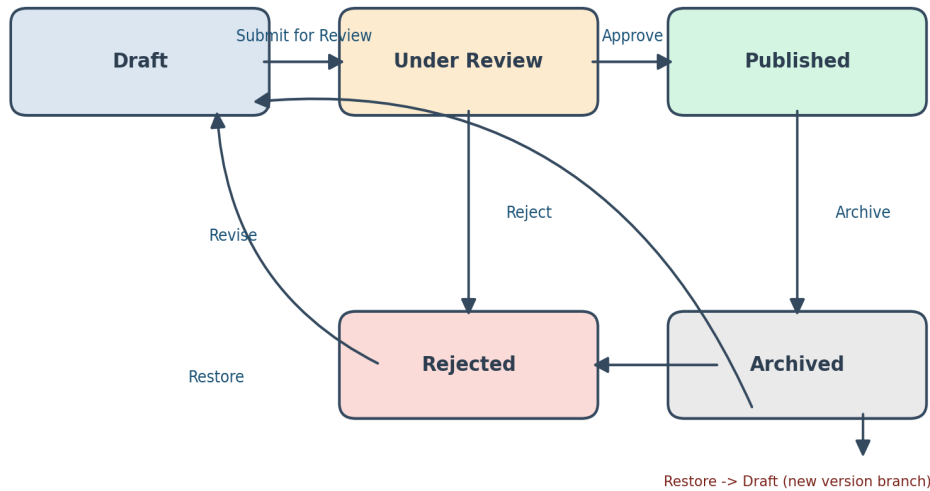
**Role-based access control involves multiple combinations of conditions (roles) and actions. A decision table enumerates every combination compactly and drives the role-permission test cases in Module D.**

| Condition / Action              | Owner | Editor | Commenter | Viewer |
|---------------------------------|-------|--------|-----------|--------|
| View document                   | Y     | Y      | Y         | Y      |
| Edit content                    | Y     | Y      | N         | N      |
| Add comments                    | Y     | Y      | Y         | N      |
| Invite / remove collaborators   | Y     | N      | N         | N      |
| Change collaborator roles       | Y     | N      | N         | N      |
| Restore previous version        | Y     | Y      | N         | N      |
| Publish / change workflow state | Y     | N      | N         | N      |
| Delete document                 | Y     | N      | N         | N      |

### 4.4 State Transition Testing

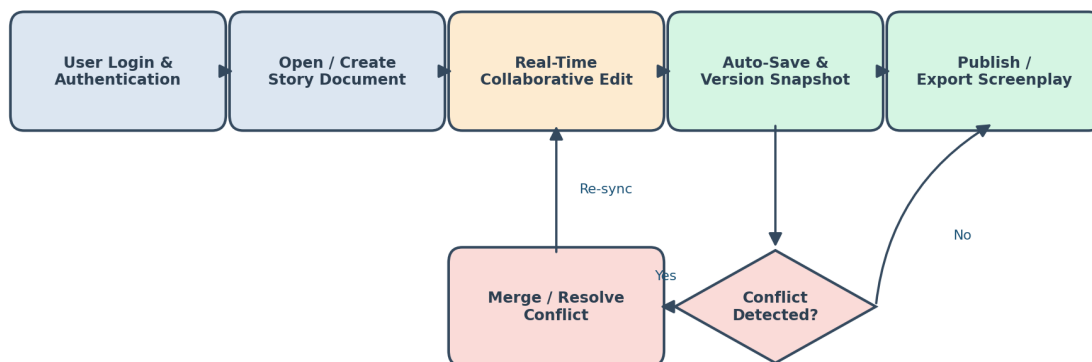
**The document workflow (Draft, Under Review, Published, Rejected, Archived) is a finite state machine. State transition testing verifies every valid transition, guards against invalid/illegal transitions, and confirms the system correctly rejects out-of-sequence events (see TC\_STATE\_04).**

| Current State | Event / Trigger   | Next State   | Guard Condition                                   |
|---------------|-------------------|--------------|---------------------------------------------------|
| Draft         | Submit for Review | Under Review | Author clicks Submit; document is non-empty       |
| Under Review  | Approve           | Published    | Reviewer has Owner/Reviewer role                  |
| Under Review  | Reject            | Rejected     | Reviewer provides rejection comments              |
| Rejected      | Revise & Resubmit | Draft        | Author edits and resubmits                        |
| Published     | Archive           | Archived     | Owner initiates archive action                    |
| Archived      | Restore           | Draft        | Owner/Editor restores; creates new version branch |

**Figure 1: State Transition Diagram - Document Version Lifecycle***Figure 1: State Transition Diagram - Document Version Lifecycle*

#### 4.5 Collaborative Workflow Overview

The diagram below illustrates the end-to-end flow exercised across Modules B, C and E - from authentication through real-time editing, auto-save/versioning, conflict resolution, and final publish/export - and shows how the test scenarios map onto the live system flow.

**Figure 2: Collaborative Editing & Version Control Workflow***Figure 2: Collaborative Editing and Version Control Workflow*

## 5. Traceability & Test Coverage

The Requirement Traceability Matrix (RTM) below maps every functional and non-functional requirement to its corresponding test scenario(s) and test case ID range, ensuring no requirement is left unverified.

| Req. ID | Requirement Summary                                                             | Test Scenario(s) | Test Case ID Range | Coverage |
|---------|---------------------------------------------------------------------------------|------------------|--------------------|----------|
| FR1     | User registration & authentication                                              | TS1, TS2         | TC_AUTH_01 - 05    | Full     |
| FR2     | Create new document                                                             | TS3              | TC_DOC_01 - 03     | Full     |
| FR3     | Real-time collaborative editing                                                 | TS4              | TC_DOC_04          | Full     |
| FR4     | Auto-save / version snapshot                                                    | TS5              | TC_DOC_05 - 06     | Full     |
| FR5     | View version history                                                            | TS6              | TC_VER_01 - 02     | Full     |
| FR6     | Restore previous version                                                        | TS7              | TC_VER_03 - 04     | Full     |
| FR7     | Assign collaborator roles                                                       | TS9              | TC_ROLE_01 - 02    | Full     |
| FR8     | Role-based action restriction                                                   | TS10, TS11       | TC_ROLE_03 - 05    | Full     |
| FR9     | Conflict detection & merge                                                      | TS12, TS13       | TC_CONF_01 - 04    | Full     |
| FR10    | Inline comments/suggestions                                                     | TS14             | TC_CMT_01 - 03     | Full     |
| FR11    | Export to PDF/FDX/Fountain                                                      | TS15, TS16       | TC_EXP_01 - 04     | Full     |
| FR12    | Notifications                                                                   | TS18             | TC_NOTIF_01 - 03   | Full     |
| FR13    | Workflow state transitions                                                      | TS17             | TC_STATE_01 - 05   | Full     |
| FR14    | Search within/across documents                                                  | TS3 (extended)   | TC_DOC_01          | Partial  |
| NFR1-9  | Performance, scalability, security, compatibility, recoverability, auditability | TS19             | TC_NFR_01 - 06     | Full     |

## 5.1 Coverage Summary

| Metric                            | Value                                                                                       |
|-----------------------------------|---------------------------------------------------------------------------------------------|
| Total Functional Requirements     | 14                                                                                          |
| Total Non-Functional Requirements | 9                                                                                           |
| Total Test Scenarios              | 19                                                                                          |
| Total Test Cases Designed         | 47                                                                                          |
| Requirement Coverage              | 100% (14/14 FR and 9/9 NFR fully or partially traced)                                       |
| Test Design Techniques Applied    | Equivalence Partitioning, Boundary Value Analysis, Decision Table, State Transition Testing |
| Condition Types Covered           | Normal, Boundary, Invalid, Exceptional                                                      |

## 6. Test Environment, Execution Strategy & Defect Classification

### 6.1 Test Environment

| Component              | Details                                                                                                         |
|------------------------|-----------------------------------------------------------------------------------------------------------------|
| Application Under Test | Collaborative Story / Screenplay Writing Platform (Web application, cloud-hosted)                               |
| Browsers               | Chrome, Firefox, Edge, Safari - latest two major versions (per NFR7)                                            |
| Devices / OS           | Windows 11, macOS, and responsive check on iOS/Android tablets                                                  |
| Backend / API Layer    | REST/WebSocket services for real-time sync, versioning and notifications                                        |
| Test Data Source       | Dedicated staging environment with seeded users, roles and sample screenplay documents                          |
| Automation Tools       | Selenium / Playwright for UI regression; Postman/Newman for API tests; JMeter for load/concurrency (NFR1, NFR2) |

| Component                | Details                                                                            |
|--------------------------|------------------------------------------------------------------------------------|
| Defect & Test Management | JIRA for defect tracking; Zephyr/TestRail for test case execution and reporting    |
| Version Control / CI-CD  | Git-based repository with automated test execution integrated into the CI pipeline |

## 6.2 Entry and Exit Criteria

### Entry Criteria

- Requirement and design documents (FR/NFR) are reviewed and baselined.
- Test environment, test data and required tool access are provisioned and verified.
- Build is deployed to the staging environment and passes a basic smoke test.

### Exit Criteria

- 100% of planned test cases (41) have been executed at least once.
- No open Critical or High severity defects remain unresolved.
- At least 95% of test cases pass, with all failures triaged and tracked to closure or an accepted risk.
- Requirement Traceability Matrix shows full coverage with no untested functional requirement.

## 6.3 Defect Severity & Priority Classification

| Severity | Definition                                                                  | Example from this Platform                                                 |
|----------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Critical | System crash, data loss, or a security breach with no workaround.           | A version restore permanently deletes newer versions instead of branching. |
| High     | Core feature is broken but the system remains partially usable.             | Real-time collaborative edits do not sync between users.                   |
| Medium   | A feature works incorrectly under specific conditions; a workaround exists. | Conflict-resolution dialog mislabels which change is 'mine' vs 'theirs'.   |
| Low      | Cosmetic, UI/UX, or minor inconsistency with no functional impact.          | Comment timestamp uses an inconsistent date format.                        |

## 6.4 Risk-Based Test Prioritization

**Test execution is sequenced by business risk so that the highest-impact areas are verified first if time is constrained, in line with a risk-based testing strategy.**

| Module                      | Risk Level | Priority | Justification                                                               |
|-----------------------------|------------|----------|-----------------------------------------------------------------------------|
| B - Real-Time Editing       | High       | P1       | Core value proposition; failure directly blocks collaborative writing.      |
| C - Version History         | High       | P1       | Data loss or incorrect restore has irreversible impact on authors' work.    |
| E - Conflict Resolution     | High       | P1       | Undetected conflicts silently corrupt shared script content.                |
| D - Roles & Permissions     | High       | P2       | Broken access control risks unauthorized edits or data exposure.            |
| J - Non-Functional/Security | High       | P2       | Security and performance issues affect trust and platform viability.        |
| A - Authentication          | Medium     | P2       | Well-understood domain with mature, reusable test patterns.                 |
| H - Workflow States         | Medium     | P2       | Incorrect transitions affect process integrity but are usually recoverable. |
| G - Export                  | Medium     | P3       | Important for interoperability but does not affect in-app data integrity.   |
| F - Comments                | Low        | P3       | Supporting feature; failure degrades collaboration but not core content.    |

| Module            | Risk Level | Priority | Justification                                                               |
|-------------------|------------|----------|-----------------------------------------------------------------------------|
| I - Notifications | Low        | P3       | Non-blocking; failure causes inconvenience rather than data or access risk. |

## 7. Documentation Notes & Conclusion

### 7.1 Assumptions

- Auto-save interval is configurable between 30 and 120 seconds; default is 60 seconds.
- Maximum of 50 concurrent collaborators and 50 retained versions per document, per NFR2 and BVA limits.
- Role hierarchy is fixed as Owner > Editor > Commenter > Viewer with no custom role creation in this scope.
- All test data shown is representative; actual execution would draw from a dedicated test environment / test data set.

### 7.2 Conclusion

**This Test Case Design document has identified the functional and non-functional requirements of the Collaborative Story/Screenplay Writing Platform with Version History, derived ten modules of test scenarios, and documented 47 detailed test cases spanning normal, boundary, invalid and exceptional conditions. Four complementary test design techniques - Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing and State Transition Testing - were applied to maximise defect-detection efficiency while minimising redundant tests. A full Requirement Traceability Matrix confirms 100% mapping between requirements, test scenarios and test cases, and a defined test environment, execution strategy and defect classification scheme round out a release-ready QA package - providing strong confidence in the correctness, reliability, security and usability of the proposed system prior to release.**

## 8. Glossary of Terms

| Term                                  | Definition                                                                                                                      |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Screenplay / Fountain                 | A plain-text markup format for writing screenplays that both humans and screenwriting software can parse.                       |
| Version Snapshot                      | An immutable, timestamped copy of a document's full content captured automatically or manually for later retrieval.             |
| Operational Transform (OT)            | An algorithmic technique that merges concurrent edits from multiple users into a single consistent document state.              |
| Conflict Resolution                   | The process, manual or automatic, of reconciling two or more overlapping edits made to the same content.                        |
| Role-Based Access Control (RBAC)      | A permissions model in which a user's allowed actions are determined by their assigned role (Owner, Editor, Commenter, Viewer). |
| Requirement Traceability Matrix (RTM) | A document mapping each requirement to the test scenarios and test cases that verify it, used to confirm coverage.              |
| Equivalence Partitioning (EP)         | A test design technique that divides input data into partitions expected to be handled the same way by the system.              |
| Boundary Value Analysis (BVA)         | A test design technique focused on values at, just below, and just above the edges of an input range.                           |
| Decision Table                        | A tabular technique that lists every combination of conditions and their corresponding expected actions or outcomes.            |
| State Transition Testing              | A technique that verifies a system's behaviour as it moves between defined states in response to events.                        |
| Entry / Exit Criteria                 | The conditions that must be met before testing begins (entry) and before testing is considered complete (exit).                 |
| .fdx (Final Draft)                    | The native XML-based file format used by the Final Draft screenwriting application.                                             |